

SATYENDRA NATH BOSE

Born in Kolkata, India, S. N. Bose wrote of the behavior of photons—particles of light—in a 1924 paper sent to Einstein, who was so impressed that he had the paper published.



Bose's (1894–1974) formulations led Einstein to predict a fifth state of matter, in addition to the known states (solid, liquid, gas, and plasma). Einstein theorized that this fifth state—named the Bose-Einstein condensate (BEC)—would form when matter possessing a dense collection of very low-energy particles—named “bosons” after Bose—is super-cooled to temperatures close to absolute zero. Scientists first made BEC particles in 1995, heralding a new field in superfluids and superconductors.

the photoelectric effect, Einstein added to the foundations of quantum theory: the area of physics that deals with light, atoms, and subatomic particles. The second paper proved molecules and atoms exist through Brownian motion.

The third paper outlined his special theory of relativity, which explained the relative motion of objects moving at constant velocity. He showed that the speed of light is constant, however fast an observer is moving relative to the light's source. But time does not run at a single speed—clocks moving relative to

“Life is like riding a bicycle. To keep your balance, **you must keep moving.**”

Albert Einstein, 1930

